



PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

Computer Networks

Assignment-01 Task#05

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DEPARTMENT	CS AND IT
PROGRAM	BS AI
SECTION	AI BLUE

For the HTTPS based website access, answer the following:

Wireshark - Packet 41 - HTTPSPcapng

```

> Frame 41: 107 bytes on wire (856 bits), 107 bytes captured (856 bits) on interface \Device\NPF_{89255AF5-6392-4079-8608-86CA571C388E}, id 0
> Ethernet II, Src: ba:7b:ef:77:20:e7 (ba:7b:ef:77:20:e7), Dst: Intel_E4:0d:5a (34:e1:2d:84:0d:5a)
> Internet Protocol Version 4, Src: 192.168.149.31, Dst: 192.168.149.45
> User Datagram Protocol, Src Port: 53, Dst Port: 54979
< Domain Name System (response)
  Transaction ID: 0x528b
  > Flags: 0x0100 Standard query response, No error
  Questions: 1
  Answer RRs: 2
  Authority RRs: 0
  Additional RRs: 0
  < Queries
    > sourceforge.net: type A, class IN
  < Answers
    > sourceforge.net: type A, class IN, addr 104.18.12.149
    > sourceforge.net: type A, class IN, addr 104.18.13.149
    [Request In: 13]
    [Time: 0.210926000 seconds]

```

```

0000 34 e1 2d 84 0d 5a ba 7b ef 77 20 e7 08 00 45 00 4---Z{ w...E-
0010 00 5d 71 a0 40 00 40 11 1d 52 c0 a8 95 1f c0 a8 .]q.@.R.....
0020 95 2d 00 35 d6 c3 00 49 47 dd 52 80 81 00 00 01 --S-I G R.....
0030 00 02 00 00 00 00 73 6f 75 72 63 65 66 6f 72 .....s ourcefor
0040 67 65 03 6e 65 74 00 00 01 00 01 c0 0c 00 01 00 ge.net.....
0050 01 00 00 01 2c 00 04 68 12 0c 95 c0 0c 00 01 00 ....h.....
0060 01 00 00 01 2c 00 04 68 12 0d 95 ....h.....

```

No: 41 · Time: 1.258379 · Source: 192.168.149.31 · Destination: 192.168.149.45 · Protocol: DNS · Length: 107 · Info: Standard query response 0x528b A sourceforge.net A 104.18.12.149 A 104.18.13.149

☒ Show packet bytes

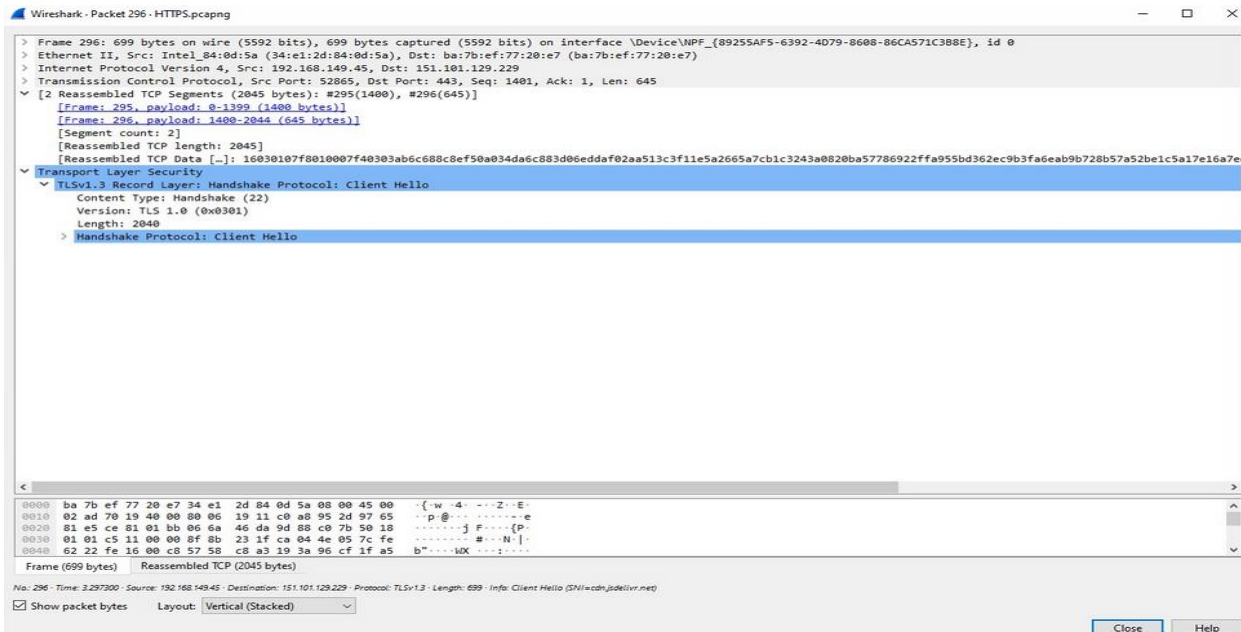
Layout: Vertical (Stacked)

Close

Help

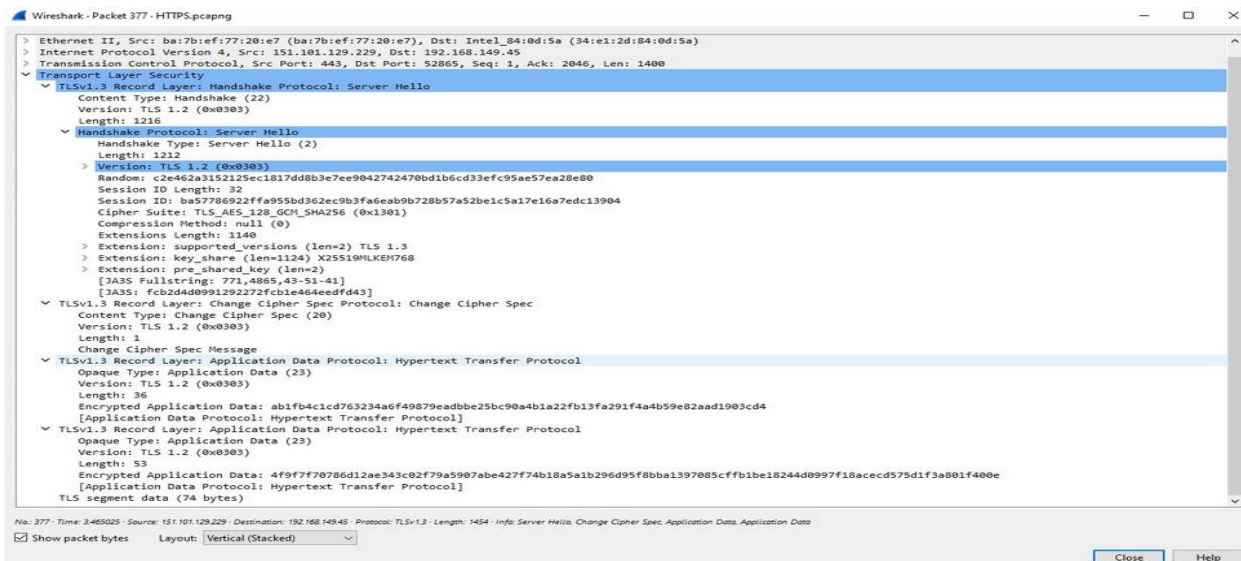
[illegible]

9. List all the TLS extensions included in the ClientHello.

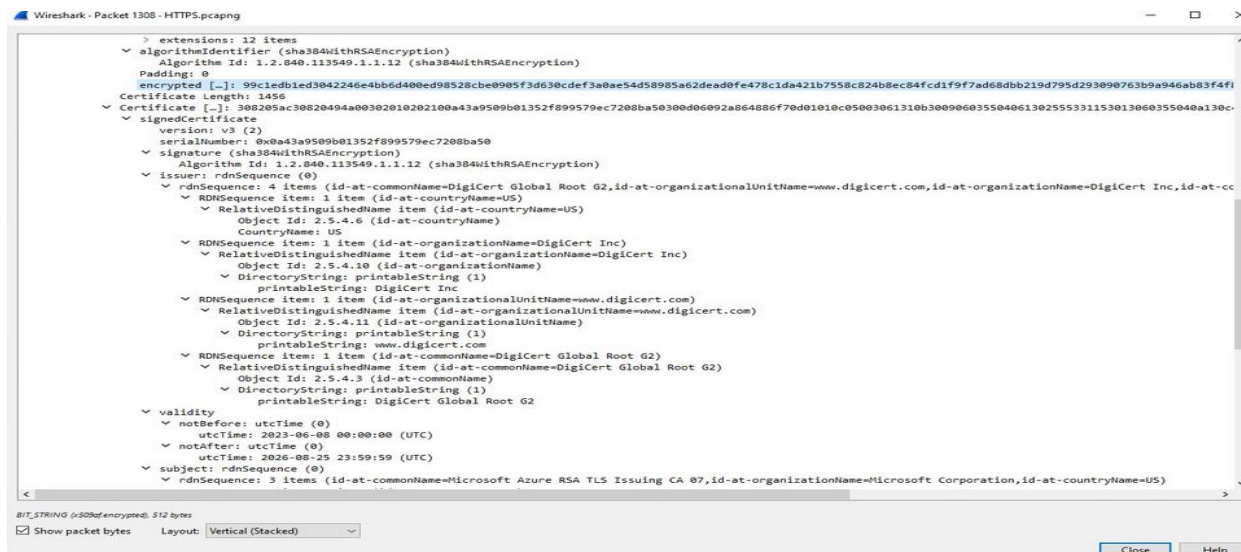


10) Identify the ServerHello message. What cipher suite is chosen by the server?

The Cipher Suite determines the algorithm used for key exchange ,symmetric encryption and message authentication.



11. Locate the Certificate message. Extract the server's certificate information (issuer, subject, validity dates)



12) After the TLS handshake, identify the first encrypted application data packet. Why can't you directly see the HTTP headers in this packet?

The HTTP headers (and the entire HTTP message) cannot be seen directly because they are encrypted by the TLS protocol.

The TLS handshake establishes a secure channel by negotiating encryption algorithms and a shared session key that is only known to the client and the server. All subsequent HTTP communication is encrypted using this key before being sent over the network as 'Application Data' packets.

Since Wireshark does not possess the private session key required for decryption, it cannot interpret the content of these packets. The payload of the 'Application Data' packet appears as opaque, random data, which effectively protects the privacy and integrity of the HTTP headers and body.

